Muhammad Najamuddin Hassan

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Education

FAST - National University of Computer and Emerging Sciences, Islamabad.

Bachelor of Science (Computer Science)	**2026**
Major: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, and Software Engineering.	
The Science School Islamabad.	**2022**
Alevel (Physics, Chemistry, Maths)	
The Science School Islamabad.	**2020**
Olevel (8 subject engineering)	

Projects

***Final Project*: VOICE — real-time Pakistani Sign Language ↔ text/speech translator**

Built FastAPI + TensorFlow/MediaPipe backend and Vite/React frontend to recognize PSL from webcam and return text/speech packaged as a full-stack final-year project.

Projects:

Edge Fall-Detection System [Python, OpenCV, Flask, YOLO, TensorRT, NVIDIA Jetson]:
Built a real-time fall detection system for edge deployment on NVIDIA Jetson using TensorRT-optimized YOLO inference. Developed a Flask-based monitoring dashboard with event logging and email alert notifications.

People IN/OUT Line-Crossing Counter [Python, OpenCV, Ultralytics YOLOv8, ByteTrack/BOTSort]:
Implemented a robust people counting system using YOLOv8 detection and tracking IDs with center/bottom-center line-crossing logic.

Custom YOLOv8 Training & Fine-Tuning Pipeline [Python, PyTorch, Ultralytics, Roboflow, Google Colab]:
Trained and fine-tuned custom YOLOv8 models using datasets exported from Roboflow. Improved model performance through dataset iteration, augmentation, and evaluation on Colab/local GPU setups.

Rubric-AI: Exam Paper OCR + Automated Evaluation Pipeline [Python, OCR, Document Parsing]:
Built an OCR-based exam evaluation workflow to extract responses from papers and apply rubric-based scoring. Generated structured outputs for review and analysis.

Trip Management System [JavaFX, OOP, GRASP & GoF Patterns]:
Developed a desktop-based trip management application using JavaFX with software design principles and patterns.

Horror Game Prototype [Unity, C#, Figma, Game UI/UX]:
Built a horror game prototype in Unity with C# scripting for player movement, environment interactions, and gameplay events. Designed interface layouts and user flow in Figma, then implemented the UI in the Unity project.

Work Experience

Co-CEO / Computer Vision Engineer, SovaVis, (sovavis.com). Co-led SovaVis operations and technical execution while building real-time computer vision **Freelance Developer (Part-Time), Fiverr / Direct Clients.** Provided code review, debugging, and project development and web technologies.	**Aug 2025 - Current**

Skills & Tools

Professional Skills	Leadership, Problem Solving, Communication, Team Collaboration, Client Handling, Project Coordination, Analytical Thinking, Time Management.
Technical Skills	Python, C++, Java, C, SQL, OpenCV, Ultralytics YOLOv8, Object Detection, Multi-Object Tracking (ByteTrack/BOTSort), TensorRT, NVIDIA Jetson, ONNX, ONNX Runtime (CUDA), Roboflow, PyTorch, Flask, Git/GitHub, HTML, CSS, WordPress, MATLAB, JavaFX, Cisco Packet Tracer, SFML.

Achievements

Awarded scholarship (no tuition fee) at college level.A Levels: 3A* O Levels: 6A*, 2A, 1B.
International Bebras Competition — Bronze Medal.
Secured positions in International Kangaroo Competitions.
Participated in IST Space Week, NSTC Maths Competition, HRCA Science Contest, and other academic competitions.

Trainings / Certification

AAttended Freelancing Seminar by Usman at IST.
Multiple certifications in AI / Data Science / Machine Learning (copies unavailable).
Practical training via hands-on project work in Computer Vision and Edge AI deployment.

Activities/ Interests

Co-Founder, Wanderlust Travel Agency: Managed 300+ clients and led operations.
Security Head, Job Fair — FAST University: Oversaw safety and logistics for 500+ organizations.
Social Initiatives: Zariya Plantation Drive, Taabeer Ration Drive; Social Media Manager at Taabeer (NGO).